Case Study 6: Comparative Analysis of Classification Algorithms for Bank Marketing Prediction

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1. Problem Statement and Objectives

A Portuguese bank sells term deposits through telephone campaigns. Most calls do not lead to a sale, so the bank wants to know in advance which customers are likely to subscribe. The problem is to build and compare classification models that predict, before a call is made, whether a customer will subscribe (y = yes), and to recommend the model that is most useful for the campaign.

The objectives were to explore and preprocess the data, implement Logistic Regression, a Decision Tree, K-Nearest Neighbours and a Random Forest on one common train/test split, compare them with accuracy, precision, recall, F1-score, ROC-AUC and confusion matrices, choose the best model using these metrics and the business cost of errors (not accuracy alone), and turn the results into recommendations for the bank.

2. Dataset Description

The data is the UCI *Bank Marketing (with social/economic context)* dataset, file bank-additional-full.csv: **41,188 rows and 21 columns**, ordered by date from May 2008 to November 2010. It was chosen over the older bank-full.csv because it is the newer UCI version and adds five social and economic context attributes, which the dataset authors found improve prediction even without call duration.

Table 1: Attribute groups (10 numerical, 10 categorical and the target)

Group	Attributes	Type
Client	age; job, marital, education, default, housing, loan	1 numeric, 6 categorical
Last contact	contact, month, day_of_week; duration	3 categorical, 1 numeric
Campaign history	campaign, pdays, previous; poutcome	3 numeric, 1 categorical
Economic context	emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed	5 numeric
Target	y: subscribed to a term deposit (yes / no)	Binary

Class balance: only **4,639 customers (11.27%) subscribed** and 36,537 (88.73%) did not, so the data is strongly imbalanced.

Data quality issues: there are no NaN values, but missing categorical information is coded as 'unknown': default 8,597 (20.87%), education 1,731 (4.20%), housing 990 (2.40%), loan 990 (2.40%), job 330 (0.80%), marital 80 (0.19%). There were **12 duplicate rows** (dropped, leaving 41,176). In pdays the value 999 means 'not previously contacted' and applies to 96.32% of customers.

3. Methodology

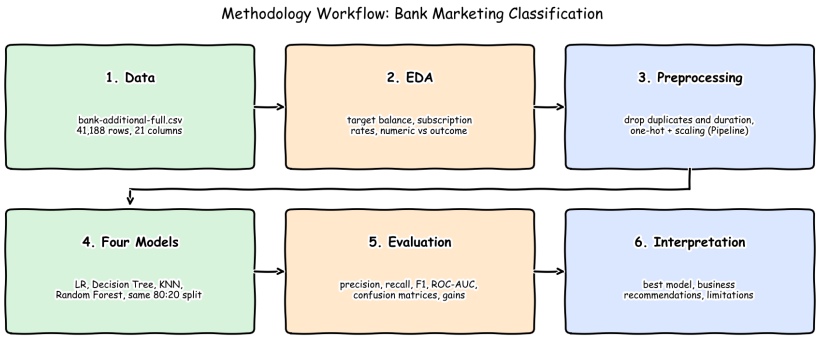


Figure 1: Methodology workflow followed in this case study

- **duration excluded:** the length of the last call is only known after the call, when the outcome is also known. The UCI notes say it should be discarded for a realistic model, so it is used in the EDA only.

- **'unknown' kept as its own category:** dropping those rows would lose over a fifth of the data, and 'unknown' behaves differently (for example, customers with unknown default status subscribe less often).
- **pdays = 999** is kept as a numeric value; after scaling it acts as a 'not contacted before' flag.
- **Encoding and scaling:** OneHotEncoder(handle_unknown='ignore') for the 10 categorical attributes and StandardScaler for the 9 numeric ones, combined in a ColumnTransformer inside each model's Pipeline, so they are fit on training data only (no leakage). y is mapped to 1 (yes) and 0 (no).
- **Split:** one stratified 80:20 split (random_state=42), reused by all four models; the test set has 928 actual subscribers.

Table 2: Models and hyperparameters

Model	Hyperparameters	Reason
Logistic Regression	max_iter=1000, class_weight='balanced'	Linear, explainable baseline; weights the rare 'yes' class
Decision Tree	max_depth=6, class_weight='balanced'	Readable rules; depth limit reduces overfitting
KNN	n_neighbors=15	Votes of 15 similar customers; has no class_weight option
Random Forest	n_estimators=300, min_samples_leaf=5, class_weight='balanced'	Ensemble of trees, captures interactions

Metrics: the positive class is 'subscribed'. **Precision** is the share of predicted subscribers who really subscribe (how many calls are not wasted); **recall** is the share of real subscribers the model finds; **F1** balances the two; **ROC-AUC** measures how well the scores rank subscribers above non-subscribers, independent of the 0.5 threshold. Accuracy is reported but not used alone, because always predicting 'no' would already give 0.8873.

4. Statistical Findings

Table 3: Summary statistics of selected numeric attributes (before removing duplicates)

Statistic	age	campaign	pdays	previous	euribor3m	nr.employed
Mean	40.02	2.57	962.48	0.17	3.62	5,167.04
Std. dev.	10.42	2.77	186.91	0.49	1.73	72.25
Min	17.00	1.00	0.00	0.00	0.63	4,963.60
Median	38.00	2.00	999.00	0.00	4.86	5,191.00
Max	98.00	56.00	999.00	7.00	5.04	5,228.10

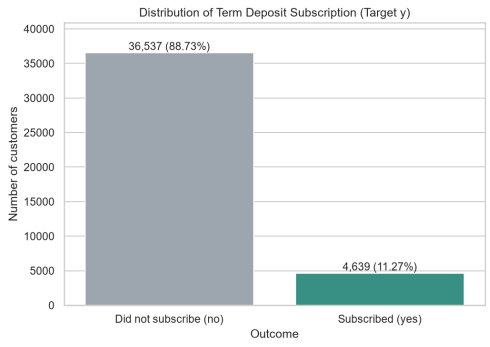


Figure 2: Distribution of the target y

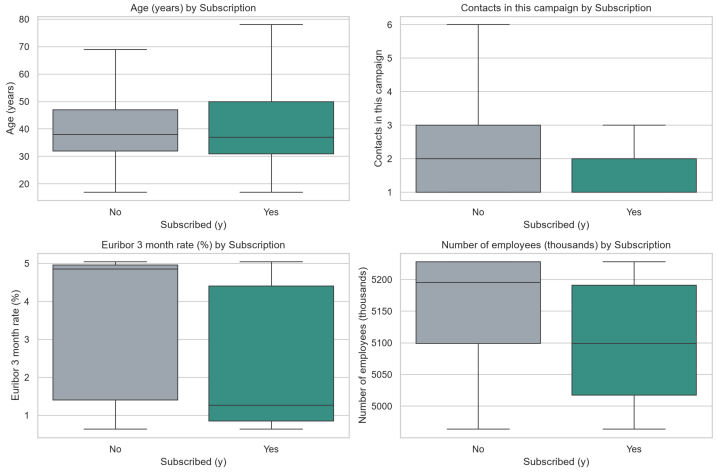


Figure 3: Numeric attributes by subscription

Only 11.27% of customers subscribed (about one call in nine). Among the numeric attributes, the economic indicators separate the classes best: subscribers have a **median Euribor 3 month rate of 1.27 against 4.86** and a lower number of employees (5,099.1 vs 5,195.8). Age barely differs (median 37 vs 38), and non-subscribers were called more often on average (2.63 vs 2.05 calls), so repeated calls do not seem to help. Call duration (EDA only) has median 449 seconds for subscribers against 164, which shows why it would leak the outcome.

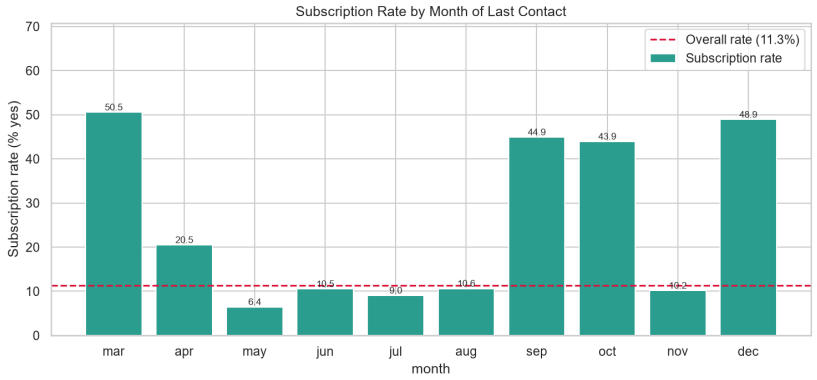


Figure 4: Subscription rate by month of last contact (dashed line: overall rate)

May had the most calls (13,767) but the **lowest rate (6.44%)**, while the quiet months had very high rates: Mar 50.55%, Dec 48.90%, Sep 44.91%, Oct 43.93%. Mass calling in busy months reaches many uninterested customers.

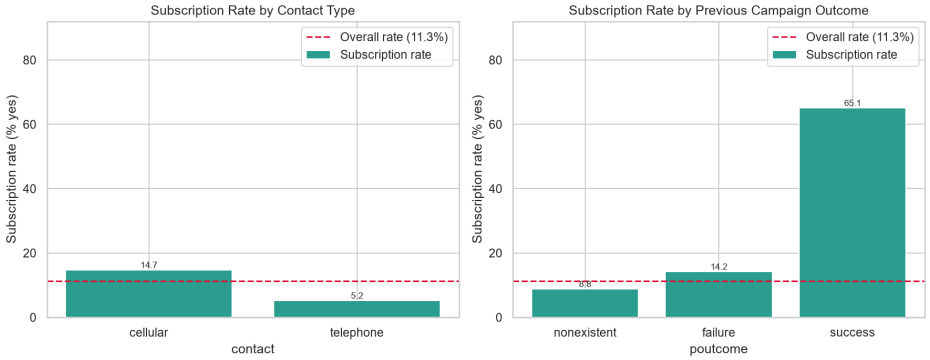


Figure 5: Subscription rate by contact type and previous campaign outcome

Customers whose **previous campaign outcome was success subscribed at 65.11%**, against 14.23% after a failure and 8.83% for customers never contacted before. Cellular contacts subscribed at 14.74% against 5.23% for landline. By job (not plotted here), students (31.43%) and retired customers (25.26%) had the highest rates and blue-collar workers the lowest (6.90%).

Evidence-based observations:

- The target is strongly imbalanced (11.27% yes), so accuracy is a weak measure.
- Previous success is the strongest single signal: 65.11% subscription, about 5.8 times the overall 11.27%.
- Economic conditions matter: subscriptions were concentrated at low Euribor rates (median 1.27 vs 4.86).
- Timing and channel matter: May 6.44% vs over 44% in the four best months; cellular 14.74% vs landline 5.23%.

5. Model Evaluation

Table 4: Final performance comparison on the test set (928 actual subscribers; positive class = yes)

Model	Accuracy	Precision	Recall	F1	ROC-AUC	TP / FN	FP	Calls / sub.
Logistic Regression	0.8303	0.3591	0.6455	0.4615	0.8002	599 / 329	1,069	2.78
Decision Tree	0.8517	0.3990	0.6239	0.4868	0.7948	579 / 349	872	2.51
KNN	0.8980	0.6106	0.2619	0.3665	0.7728	243 / 685	155	1.64
Random Forest	0.8571	0.4129	0.6358	0.5006	0.8080	590 / 338	839	2.42

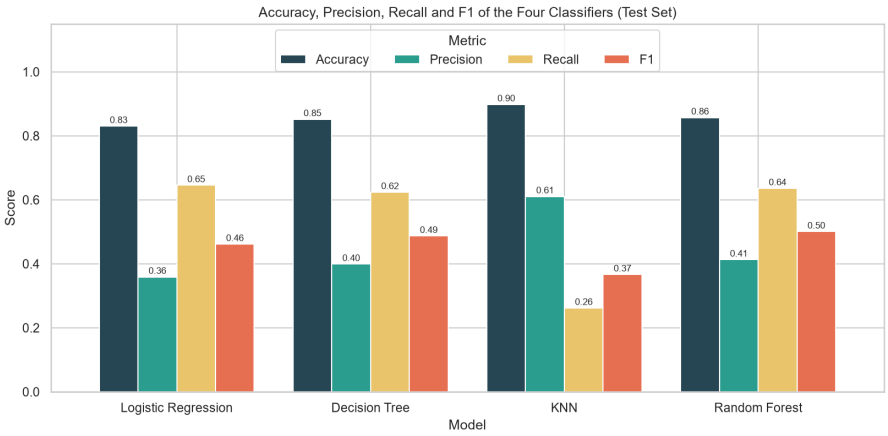


Figure 6: Accuracy, precision, recall and F1 of the four classifiers

Accuracy hardly separates the models (0.83 to 0.90) and three of them are below the 0.8873 of always predicting 'no'. Recall and F1 tell a different story: **KNN has the highest accuracy (0.8980) and precision (0.6106) but finds only 243 of 928 subscribers (recall 0.2619)**, because it has no class weights and most neighbours of any customer are non-subscribers. The three balanced models all find about 62% to 65% of subscribers.

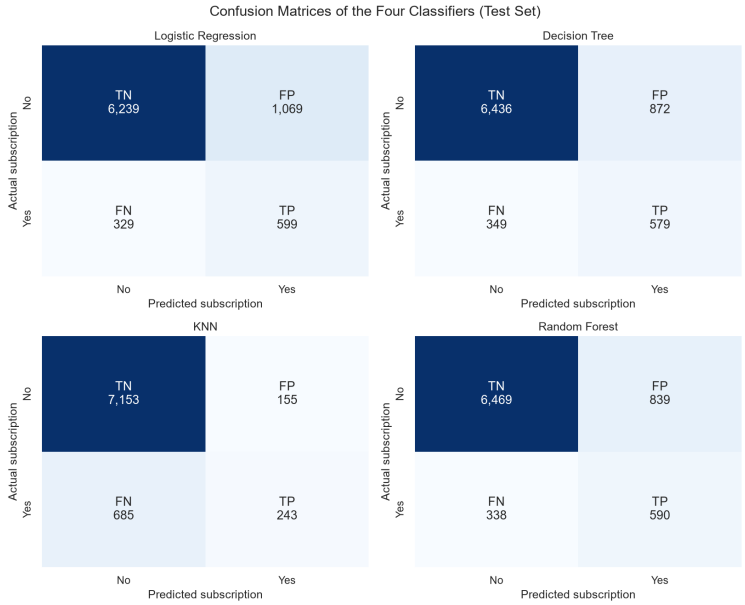


Figure 7: Confusion matrices of the four classifiers (TN, FP, FN, TP)

In practice a **false negative** is a customer who would have subscribed but is never called, a lost deposit; a **false positive** is a call to a customer who says no, a few minutes of an agent's time. Logistic Regression misses the fewest subscribers (329) but wastes the most calls (1,069). Random Forest misses only 9 more

(338) while making 230 fewer wasted calls (839). KNN wastes only 155 calls but misses 685 subscribers, the most expensive kind of error. For every model false positives outnumber true positives, the price of finding rare subscribers.

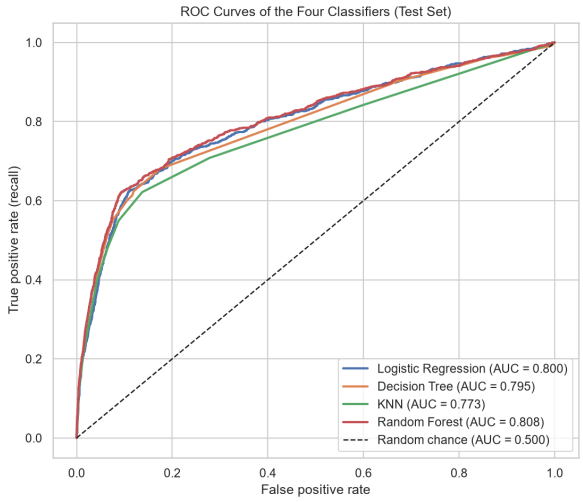


Figure 8: ROC curves with AUC

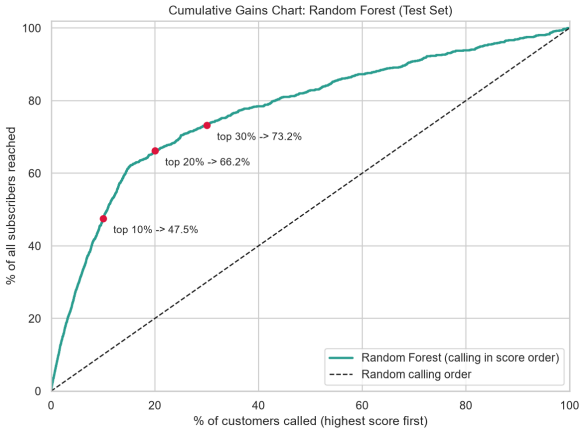


Figure 9: Cumulative gains of Random Forest

All ROC curves are well above the diagonal; Random Forest has the highest AUC (0.8080), just ahead of Logistic Regression (0.8002). The gains chart shows the value of ranking: calling the top 10% of customers by Random Forest score reaches **47.52%** of all subscribers (lift 4.75), the top 20% reaches 66.16% and the top 30% reaches 73.17%.

6. Interpretation and Business Recommendations

Best model

Random Forest is the best model. It has the best F1 (0.5006) and ROC-AUC (0.8080), recall of 0.6358 (590 of 928 subscribers) and the best precision of the three balanced models (0.4129), needing 2.42 calls per subscriber found against 2.78 for Logistic Regression. Since missing a subscriber costs more than one extra call, KNN is rejected despite its accuracy. Random Forest suits the bank because it handles the imbalance with class weights, uses only information available before the call, captures interactions between economic conditions, timing and customer history, and produces scores that rank customers well. Logistic Regression is a close second and would be preferred if a simpler, fully explainable model were required.

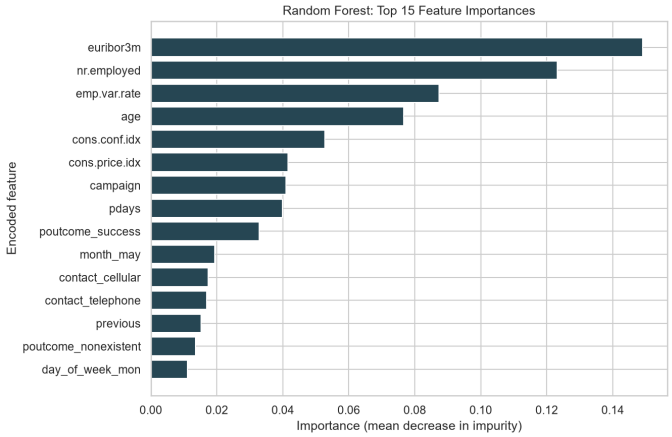


Figure 10: Random Forest: top 15 feature importances

The most important features are the economic indicators euribor3m (0.1491), nr.employed (0.1231), emp.var.rate (0.0874), followed by age, cons.conf.idx, cons.price.idx. Campaign history (campaign, pdays,

poutcome_success) and the contact month and type also appear in the top 15, in agreement with the EDA. Impurity-based importance favours continuous features, so binary columns such as poutcome_success may be under-rated.

Business recommendations

- **Call customers in score order.** Rank customers by the Random Forest score and call from the top. On the test set the top 20% of the list held 66.16% of all subscribers, so the bank could reach about two thirds of potential subscribers with one fifth of the calls, and set the cut-off by agent capacity.
- **Prioritise previous successes and mobile numbers.** Previous-success customers subscribed at 65.11% and cellular contacts at 14.74%.
- **Avoid repeat calls and over-concentration in one month.** May produced 13,767 calls at 6.44%; spreading calls and capping repeat attempts would save effort.
- **Time campaigns with economic conditions.** Euribor and employment indicators were the strongest predictors, so campaign size can be raised when these are favourable.

7. Conclusion and Limitations

On 41,176 customers with only 11.27% subscribers, four classifiers were compared on one leak-free pipeline without call duration. **Random Forest** gave the best balance (F1 0.5006, ROC-AUC 0.8080, recall 0.6358, precision 0.4129) and, used as a ranking tool, reaches 73.17% of subscribers with 30% of the calls. KNN showed how misleading accuracy is on imbalanced data.

- **2008 to 2010 data:** the data comes from one bank during the financial crisis and the top features are economic indicators, so the model must be retrained if conditions change.
- **duration excluded:** correct for pre-call use, but it removes the strongest signal, which limits F1 to 0.5006.
- **Default 0.5 threshold:** precision and recall depend on the cut-off (KNN's low recall is largely a threshold effect); the bank should choose it from call capacity.
- **Limited tuning and one split; 'unknown' values:** hyperparameters were not searched, and 20.87% of default values are unknown, which adds noise.

8. References

- S. Moro, P. Cortez and P. Rita. A Data-Driven Approach to Predict the Success of Bank Telemarketing. Decision Support Systems (2014), doi:10.1016/j.dss.2014.03.001.
- Bank Marketing dataset, UCI Machine Learning Repository: http://archive.ics.uci.edu/ml/datasets/Bank+Marketing (social and economic attributes published by Banco de Portugal).
- Libraries: pandas 3.1.0, numpy 2.5.4, matplotlib 3.11.2, seaborn 0.13.2, scikit-learn 1.9.1, reportlab 5.1.0.
- GitHub repository: <GITHUB REPO URL>

9. AI Assistance Acknowledgement

Code structure drafted with AI assistance (Claude); executed, verified and interpreted by Nikhil Kotte. Dataset sources: Bank Marketing (with social/economic context), UCI Machine Learning Repository, bank-additional-full.csv; Moro, Cortez and Rita (2014).